

Soft Sensors Based on Deep Neural Networks for Applications in Security and Safety

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Abstract—Here, this article reports about the design of a soft sensor (SS) able to monitor the hazardous gases in industrial plants. The SS is designed to estimate the gas concentrations by means of the measurements coming from an array of sensors, avoiding at the same time the humidity and temperature influence on array outputs. The SS has been designed with a data-driven approach, using a set of experimental data acquired in a laboratory. The design methodologies of two different SSs are compared, with the aim of obtaining both the good performance and a low computational complexity. As a first approach, a principal component analysis (PCA) has been performed to exploit the high correlation among some of the measures coming from the sensor array. A classical multilayer perceptron neural network is then trained to estimate the relationships between the PCA outputs and the gas concentrations. As a second approach, a deep belief network (DBN) has been considered. The data here reported show a good accuracy in the evaluation of several gas concentrations, even in the presence of noised measurements, allowing an efficient risk warning. Even if both the methods gave a similar performance, a lower number of model parameters and a lower noise sensitivity are obtained when using DBNs.

Index Terms—Deep neural networks, environmental monitoring, gas sensing systems, nonlinear system identification, principal component analysis (PCA), safety monitoring, soft sensing.

I. INTRODUCTION

IN RECENT years, many research activities have been focused on the design and validation of monitoring systems for environmental conditions, aimed at the detection of possible harmful conditions needing a proper solution [1], [2].

Among these, soft sensors (SSs) are becoming a reliable solution in many applications regarding several industrial fields such as chemical plants, refineries, and urban pollution monitoring. Furthermore, they can be employed to approach several problems in a range spanning from risk detection, what-if analysis, measuring system backup, real-time prediction for plant control to fault diagnosis strategies, and sensor/actuator validation [3]–[13].

Manuscript received July 24, 2019; revised February 5, 2020; accepted March 9, 2020. Date of publication April 7, 2020; date of current version September 15, 2020. The Associate Editor coordinating the review process was Bruno Ando. (Corresponding author: Nicola Donato.)

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Digital Object Identifier 10.1109/TIM.2020.2984465

An important role in industrial plant monitoring is played by gas monitoring systems, designed for the detection of dangerous gases, to avoid harmful conditions to the plant operators and environment and the damaging of the plant itself [3], [15]. A typical system for gas monitoring usually consists of a gas sensor arrays hosted inside the plant. In many applications, the array is composed of devices with different transduction mechanisms. In order to monitor the gas presence and the related concentration and to take the necessary action, a further processing of the sensor outputs is needed. Several algorithms can be employed for both detection and estimation of the gas presence and related concentrations, on the basis of measurements coming from gas sensors. Among them, artificial neural networks (ANNs) and fuzzy systems [15]–[18] have shown a high efficiency. By considering that a change in the environmental conditions (temperature and humidity) can influence the sensor performance, ANNs have been applied in order to compensate the sensor outputs from temperature and/or humidity effects [19]–[24]. In [20] and [21], this kind of compensation is performed on the outputs of specific gas sensors related to concentration variations. In [22]–[24], ANNs are employed for the evaluation versus temperature and/or humidity of a single gas concentration. In [25], a multilayer perceptron (MLP) is employed to evaluate, at the same time, concentration values of several gases by means of an array of sensors and by the evaluation of the environment temperature and humidity effects, then performing compensation of the entire monitoring system.

Here, the neural model identification procedure has been redesigned, with the aim to enhance the SS performance and to reduce the number of model parameters. Two different approaches are compared. In the first approach, the high correlation between some inputs is exploited by using principal component analysis (PCA), to reduce both the number of the input variables and that of the model parameters and to increase model reliability. In the second approach, a deep belief network (DBN) is used [26]–[28]. The employment of a DBN can, in fact, lead to a more compact model representation with respect to a shallow neural model, reducing also the generalization error. By looking at this article organization, Section II describes the gas monitoring system. The SS design is discussed in Section III. Section III-A reports the results obtained with the PCA and MLP, whereas the DBN approach is presented in Section III-B along with the obtained results.

Section IV contains a comparison of the results obtained with the two methods and their discussion. The robustness of the SSs to noisy measurements is also evaluated through suitable simulation with a different level of noise.

II. MONITORING SYSTEM DESCRIPTION

Five of the sensors composing the array are made by FIGARO USA Inc., Arlington Heights, IL, USA. They can be classified by the transduction mechanism:

- 1) semiconductor type: TGS-825 (hydrogen sulfide sensor) and TGS-2611(hydrogen and hydrocarbons sensor);
- 2) catalytic type: TGS-6810 (methane and LP gas sensors) and TGS-6812 [methane, hydrogen, and light petroleum gas (LPG) sensors];
- 3) electrochemical type: KE-50 (oxygen sensor).

In order to evaluate the environmental condition effects on sensor outputs, the array is completed with an HTG-3535 temperature/humidity sensor (Measurement Specialties Inc., USA) and an XFAM pressure sensor (Fujikura Ltd., Japan). A custom hardware interface has been developed for the sensor biasing and signal conditioning. A NI DAQ-6008 USB board (National Instruments, USA) equipped with a homemade software interface has been employed for the acquisition of the signals coming from the sensors.

Fig. 1 shows the functional diagram of the e-nose, and it can be seen that the sensors are not selective for a specific gas target, but three of them can be employed in monitoring methane (TGS6810, TGS6812, and TGS 2611) and two of them can be employed for LPG (TGS6810 and TGS6812). Of course, the sensors were chosen taking into account features in monitoring hexane and pentane and the concentration ranges. In such a context, the system can be considered more robust in the monitoring of these specific gases. The sensor array is mounted in a stainless chamber, to expose the device to a controlled atmosphere during the characterization procedure. By connecting it with the gas system, as shown in Fig. 2, it is possible to expose the sensors to a constant mixture flux (100 cc/min) and to switch between the reference gas value and the gas target one. The gas target concentration values are provided by the certified bottles and set by the mass flow controllers, with exception of the H_2S value, achieved by means of a permeation tube, and the humidity one, provided by a water bubbler in a thermostatic bath. The characterization setup is automated by a homemade software, allowing to reach several operating conditions and to choose the parameter subject of variation.

III. SS DESIGN

A SS can be defined as a model of a system, which is used to estimate some hard-to-measure variables based on a set of easy-to-measure ones [3]. Due to the complexity of industrial systems, such models are generally nonlinear.

They are often designed using data-driven approaches, starting from the suitable sets of experimental data. Here, an SS able to evaluate the concentration values of six gases (O_2 , H_2 , CH_4 , H_2S , C_5H_{12} , and C_6H_{14}) has been proposed.

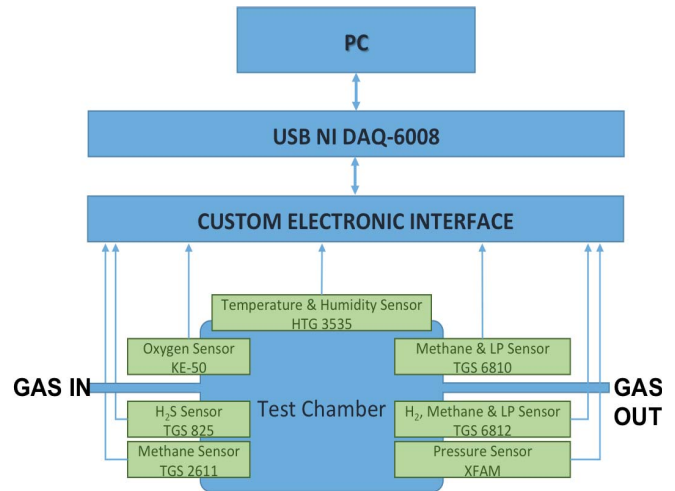


Fig. 1. e-nose development diagram.

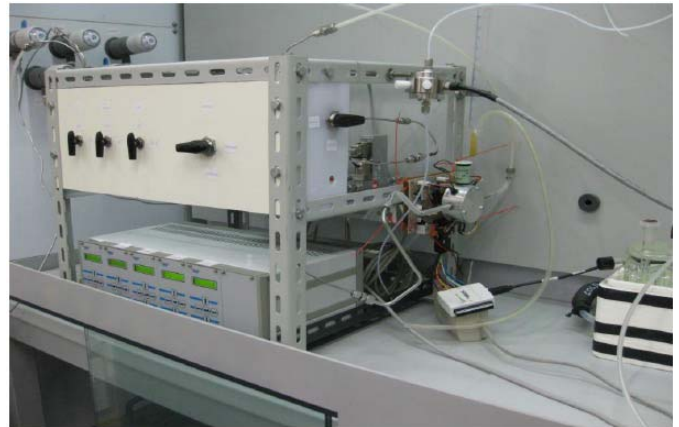


Fig. 2. Characterization setup.

A preliminary approach to this problem is proposed in [25], where the design of a model based on a MLP, trained to predict the concentration values of the six gases on the basis of the sensors array described in Section II, has been described. Considering that the sensors' response is affected by the environmental temperature and humidity, their values, in terms of voltage signals, are measured and included in the input data set. Therefore, the SS inputs are the outputs of five gas sensors (TGS825, TGS6810, TGS6812, TGS2611, and KE50), temperature, and humidity values. The pressure is not considered as an input because its value is kept constant during the entire monitoring process.

In this article, two data-driven approaches to the SS design are proposed. They have been developed using an experimental data set acquired with the characterization apparatus described in Section II. A number of 55 values of gas concentrations at several temperature/humidity conditions were recorded, for a total amount of 1028 measurements. Then, the training set was created, by selecting 900 measured samples. The test data set contains, therefore, 128 samples. The training data set was selected in order to cover at best the input space though maintaining diversity with respect to the test data set [29].

TABLE I
GAS CONCENTRATION RANGES

Output name	Gas Name	Concentration range (ppm)
y1	Oxygen	0-200000 (0%-20%)
y2	Hydrogen	0-30000
y3	Methane	0-10000
y4	H ₂ S	0-19
y5	Hexane	0-11840
y6	Pentane	0-13840

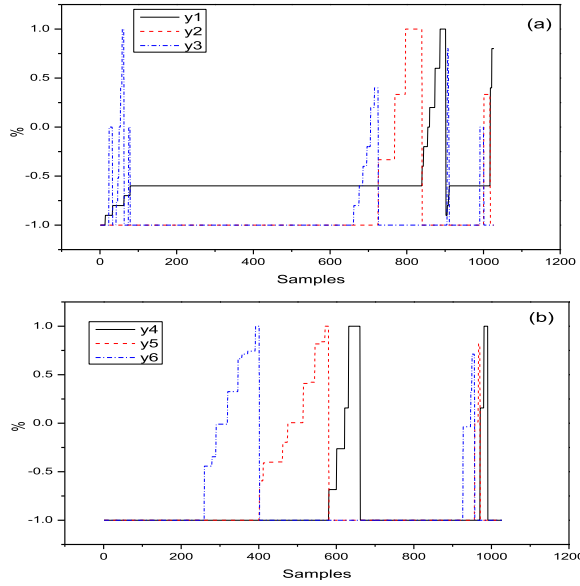


Fig. 3. (a) and (b) SS outputs data set.

TABLE II
LIST OF THE SS INPUTS

Input name	Sensor Name
u1	TGS6810
u2	TGS825
u3	TGS2611
u4	KE50
u5	TGS6812
u6	Humidity
u7	Temperature

Furthermore, the training data set as been further randomly split into two data sets: training (80%) and validation (20%) ones, to apply cross-validation during the learning phase.

The concentration ranges for each gas (SS outputs) are shown in Table I (for a better readability, the oxygen concentration range is also reported in percentage). The relative humidity value and temperature one were set in the range of 16%–40.5% and 24 °C–37 °C, respectively. The experimental data are shown in Figs. 3 and 4. The input nomenclature is shown in Table II.

As it can be observed from Fig. 4, some of the input variables are highly correlated, in particular, those coming

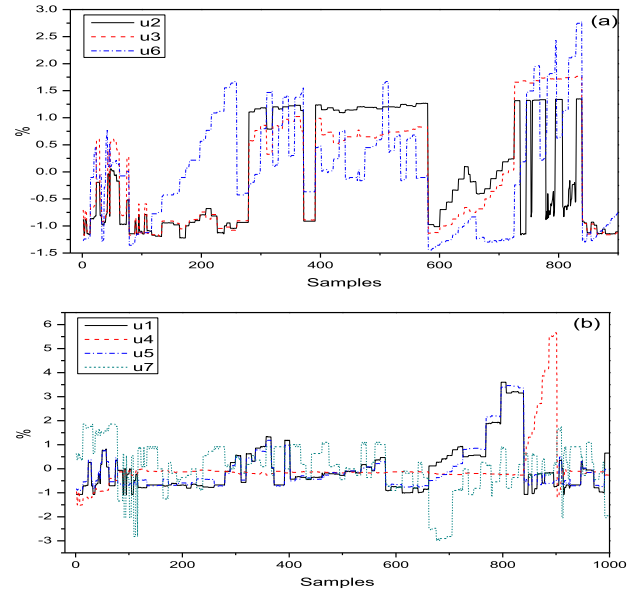


Fig. 4. (a) and (b) Input data set.

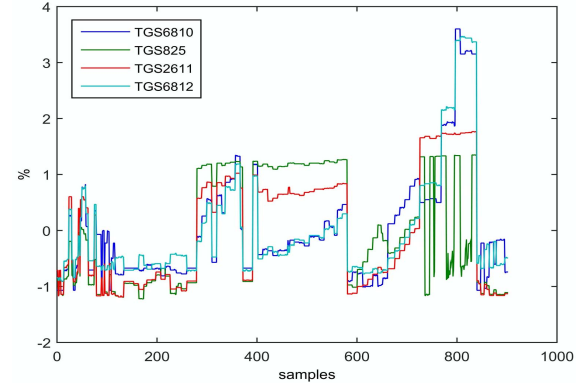


Fig. 5. Subset of the input variables.

from the TGS sensors (u1—TGS6810, u2—TGS825, u3—TGS2611, and u5—TGS6812). This is better shown in Fig. 5, where only these four inputs are represented. To exploit this correlation, as the first approach, PCA is used before training the neural model, as described in Section III-A.

The following indexes have been used to evaluate the SS performance.

Maximum Absolute Error (MAE):

$$\text{MAE} = \max_{i=1}^N (|Y_i - T_i|). \quad (1)$$

Pearson–Product–Moment correlation coefficient (CC):

$$\text{CC} = \frac{\sum_{i=1}^N |Y_i - \bar{Y}| \cdot |T_i - \bar{T}|}{\sqrt{\sum_{i=1}^N (Y_i - \bar{Y})^2 \cdot (T_i - \bar{T})^2}} \quad (2)$$

where Y_i is the corresponding model output to the i th input sample, the target response is T_i , and N is the total number of samples.

A. SS Based on PCA and MLP

PCA and partial least-squares (PLS) regression are reliable tools [30], especially in the SS design. They are employed

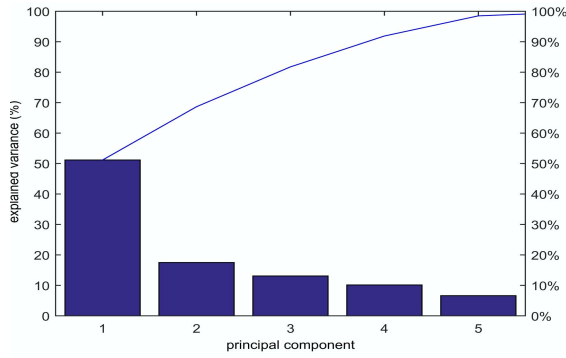


Fig. 6. Pareto chart of the variance explained by the first five principal components.

in several stages of a typical design procedure, ranging from outlier detection to model identification [3]. PCA can be viewed as a mean of compressing data. More interestingly, as reported in [29], it can be also viewed as an unsupervised learning algorithm that learns a representation of the input data having a lower dimension.

In this new representation, the input variables have no linear correlation with each other. This can be considered as a first step to achieve a representation, in which the elements are statistically independent. To obtain the full statistical independence, the algorithm should also remove any nonlinear relationship between variables.

The linear projection performed by the PCA algorithms aligns the direction of the greatest variance of the original data set X with the axes of the new space Z . The resulting representation has a diagonal covariance matrix, which implies that the individual elements of Z are mutually uncorrelated.

To apply PCA to the whole set of input data, they are first normalized in order to obtain zero mean and unit variance. The normalized data are then organized in a matrix X , which contains a number of columns equal to the number of inputs, and the covariance matrix is computed as follows:

$$\text{COV}_X = X^T X. \quad (3)$$

The eigenvectors p_i of this matrix (loadings) identify the directions, where the majority of data variability occurs.

The corresponding eigenvalues, giving the amount of variability associated with each direction, are in the application: $\text{PCA_latent} = [3.58 \ 1.22 \ 0.91 \ 0.70 \ 0.46 \ 0.08 \ 0.02]$. As it can be observed from the values of PCA_latent and from the corresponding Pareto chart in Fig. 6, representing the percentage of variance explained by each principal component, the first five components are sufficient to represent almost 100% of the data variance and can be used as SS inputs. It can also be observed that the first four PCs explain 95% of the variance; therefore, a model with four inputs is also worth to be investigated. This, however, did not give satisfactory results, and only, the results obtained by using five PCs are reported in the following.

After PCA projection, a model with five inputs is implemented by using an MLP, as shown in Fig. 7.

An MLP with five input units and six outputs, relating the gas concentrations and the PCA output variables, has

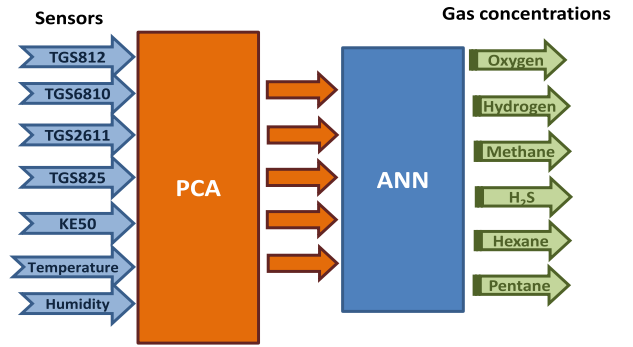


Fig. 7. Proposed SS for the gas concentration evaluation.

been trained by employing the Levenberg–Marquardt training algorithm [18], with early stopping procedure, in order to avoid overfitting. The MATLAB implementation of the training algorithm (trainlm, toolbox neural networks) was employed with the default values of the parameters. To implement the early stopping, during the learning phase, the learning data set has been, therefore, further split. About 80% of the data were randomly selected to perform the learning, whereas the remaining 20% to evaluate the performance. The test data set has been used only to compare the performance of the two proposed SS design methodologies. A growing strategy was used to select the network topology; a maximum number of two layers have been allowed, with a number of neurons going from 0 to 20 for each layer. The best performance was obtained with a network topology made by 5–12–12–6 neurons, corresponding to 355 model parameters (weights, biases, and PCA parameters).

About 30% less free parameters are used with respect to the ANN without PCA reported in [25]. This is a good result in view of a future implementation on an embedded system.

The values of the indexes (1) and (3), computed both on the training and test data sets, are reported in Section III-B, in comparison with the results obtained with DBN. In Fig. 8, a comparison between the targets and the estimated output values on the whole data set is reported. In Fig. 8, the first 900 data have been used during the learning phase, whereas the last 128 belong to the test data set. The corresponding residuals and the histograms of the residuals are shown in Figs. 9 and 10, respectively.

In Figs. 9 and 10, the obtained performance can be considered to be very satisfactory. The number of parameters of the model is, however, still high. In Section III-B, a different approach is used to design the SS, exploiting the use of DBNs to further reduce model complexity.

B. DBN Approach

As already discussed, PCA can be viewed as a linear feature extraction algorithm, retaining the PC which explains most of the data variance.

Deep neural networks, and in particular, DBNs can also be considered a way to extract essential feature from data [26], each layer extracting some feature from its input up to the disentanglement of complex features. They are called deep because the number of layers (depth) is more with respect to

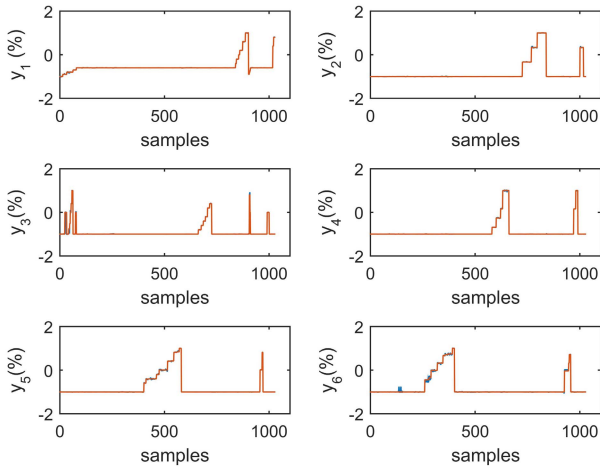


Fig. 8. Normalized target (red) and the normalized estimated outputs (blue).

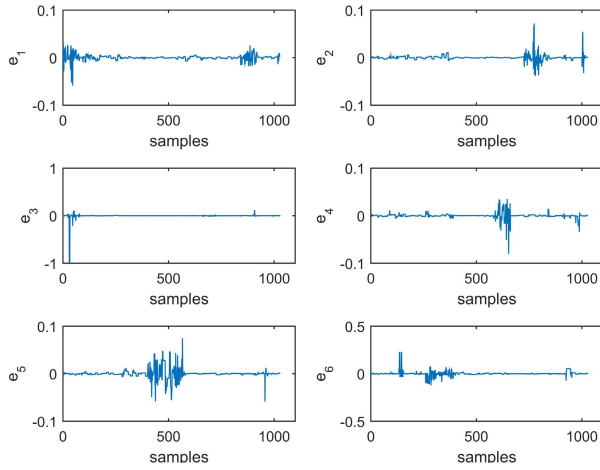


Fig. 9. Residual between the target and the estimated outputs.

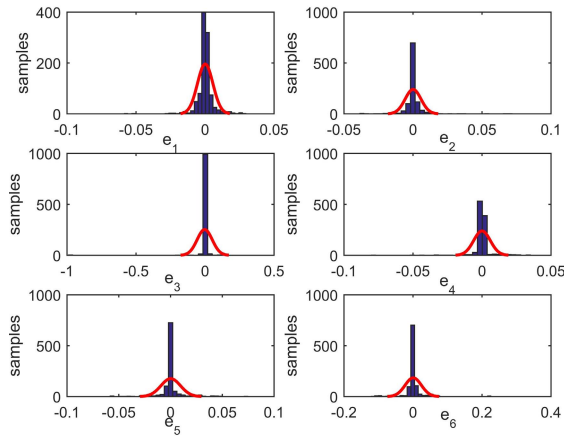


Fig. 10. Histogram of the residual between the target and the estimated outputs.

the conventional one-hidden-layer MLP, which is, in contrast, called as a shallow network. In detail, a network with up to three (two hidden) layers is called shallow, whereas those with more than three are called deep.

A shallow MPL has been widely used in industrial applications, due to their universal approximation capability and efficient learning algorithms.

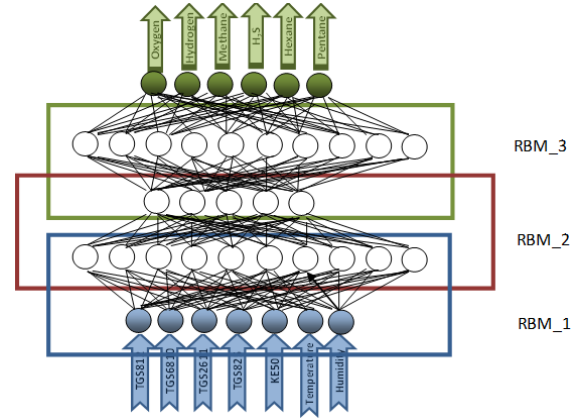


Fig. 11. Scheme of the deep network used to develop the SS.

In some applications, however, the high number of hidden neurons required by the network creates problems during learning, leading to overfitting and models with a high number of parameters. The deep networks, in contrast, often allow obtaining more efficient and compact models.

Most of the applications of deep learning are in the fields of object recognition, speech recognition, language processing, and medical science [26]. In this article, they are used to efficiently approach the SS design.

The back propagation (BP)-based training algorithms fail to provide a satisfactory performance when used for deep networks, mainly due to the problem of vanishing gradients [26]. Suitable learning algorithms should, therefore, be used. A new training method for deep networks is described in [31].

By means of an unsupervised learning algorithm, the pre-training of each layer is performed, one by one. The weights obtained at the end of the unsupervised learning are then used as initial weights to perform a fine-tuning of the network, by using a supervised algorithm.

Among the different structures of deep networks, a DBN is used in this application [26], [32]. It is composed by a stack of restricted Boltzmann machines (RBMs). Each RBM is composed of two layers. The lower visible layer v , for the inputs, and the hidden one h give the latent variable representation.

The hidden layer of each RBM is the visible input layer of the next RBM of the stack. The outputs of the last RBM of the stack are then connected, by using another set of weights, to the output units. Fig. 11 shows the block diagram of the stacked RBM, containing three RBMs and six output neurons, employed in this work.

As discussed above, the training of a stack of RBMs is made up of two steps. During the first step, a greedy layer-wise unsupervised learning algorithm, from the low layer to the higher one, is used. It can be done by maximizing the likelihood of the observed samples of the input and considering the variables associated with the first layer as hidden ones, by using the contrastive divergence algorithm [28], [30], [31]. Once the weights of the first layer are determined, the outputs of this layer are used as visible inputs of the next RBM.

This first unsupervised training step generates the latent variables at the topmost layer. Such variables need to be further

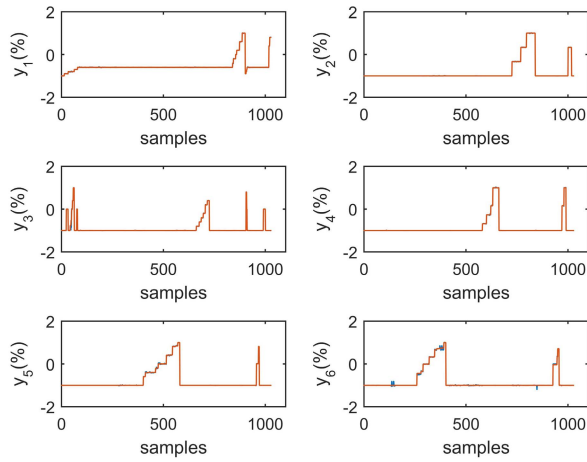


Fig. 12. Comparison between the normalized target (red) and the normalized estimated outputs (blue).

processed for estimating the DBN output. In the second phase, an output layer is added, whose weights are randomly initialized and the network is fine-tuned by a BP-like supervised learning algorithm, starting from the initial weight configuration settled by the unsupervised phase. Again, the Levenberg–Marquardt algorithm has been used, with the same approach of Section III-A, by randomly splitting the data (80% and 20%) and by implementing the early stopping procedure.

The networks considered in this work were trained with the MATLAB DeeBNetV3.2 toolbox [32].

To find the network topology, all the possible combinations have been used, setting the number of layer to 3 and allowing a maximum of 20 neurons for each layer. The best results have been obtained with the network with 7–10–5–10–6 units. This model has, therefore, a total number of 285 parameters, which is quite lower with respect to the previous approaches. The results obtained on the whole data set are shown in Figs. 12–14. Again, the first 900 data have been used during learning, whereas the last 128 are fresh data. In Figs. 12–14, a very good performance of the method can be appreciated, considering, in particular, the low ranges of the residuals and the corresponding histograms. A deeper analysis of the results is reported in Section IV, in comparison with the results obtained by using PCA and a shallow MLP.

IV. MODEL COMPARISON

In Table III, a comparison of the performance obtained with the two proposed methods is reported. For each of the six estimated outputs, the root-mean-square errors (RMSEs) and the CC were computed as defined in (2), and the MAE was computed as defined in (1), on the learning data set (RMLS_l, CC_l, and MAE_l) and on the test data set (RMLS_t, CC_t, and MAE_t) to appreciate the generalization capability of the models.

Both the methods allow accomplishing very well the desired gas evaluations. Slightly better results, in particular, with regard to the generalization ability, are obtained by using DBNs.

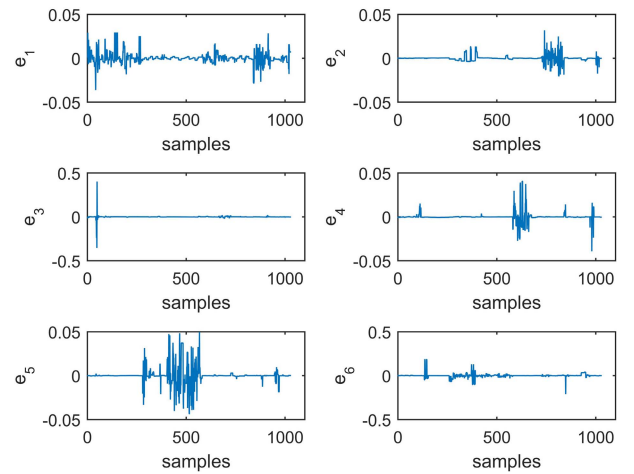


Fig. 13. Residual between the target and the estimated outputs.

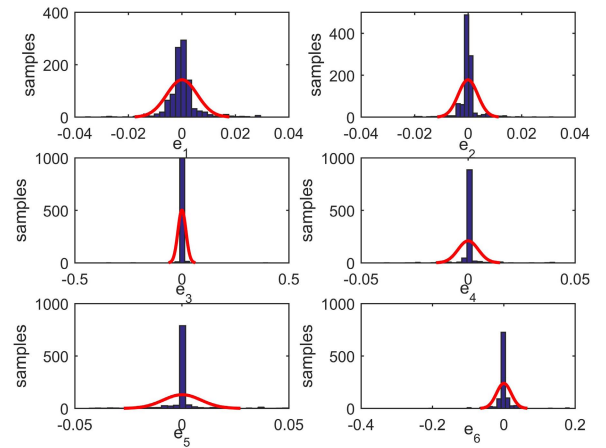


Fig. 14. Histogram of the residual between the target and the estimated outputs.

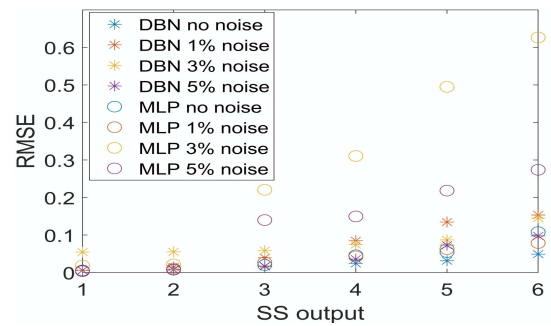


Fig. 15. RMSE in the presence of noise at a different percentage.

Moreover, taking into account also model complexity, i.e., the number of parameters of the model, the second approach is much more advantageous: the number of parameters was reduced from 456 (the simple MLP reported in [30]) to 355 (PCA + MLP) and finally to 285 (deep MLP). Taking into account these considerations, it would be possible to develop a compact analyzer system in an embedded board.

To evaluate the robustness of the SSs to noisy measurements, different levels of noise have been added. By considering the different transduction mechanisms of the employed

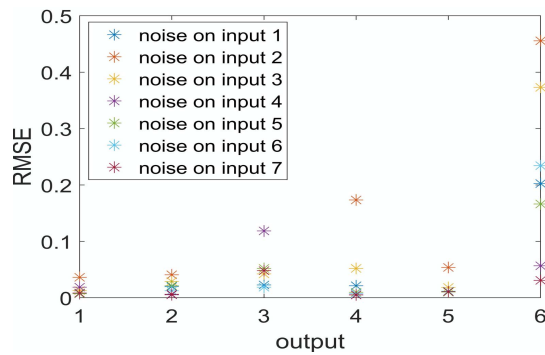


Fig. 16. RMSE in the presence of 5% noise added at one input at the time for the model implemented by the PCA + MLP.

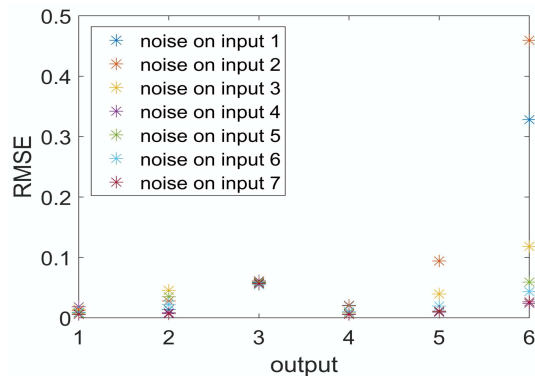


Fig. 17. RMSE in the presence of 5% noise added at one input at the time for the model implemented by the DBN.

TABLE III
COMPARISON OF THE RESULTS

Output	Model	RMSE _l	RMS _t	CC _l	CC _t	MAE _l	MAE _t
1	MLP	0.0013	4.64e-04	0.9998	0.9999	0.0586	0.0187
	DBN	0.0014	5.00e-04	0.9998	0.9999	0.0361	0.0284
2	MLP	0.0013	6.54e-04	0.9999	0.9999	0.0720	0.0545
	DBN	8.98e-04	2.22e-04	1.0000	1.0000	0.0318	0.0174
3	MLP	0.0139	0.0011	0.9818	0.9996	0.9913	0.1197
	DBN	0.0047	2.50e-04	0.9979	1.0000	0.4024	0.0145
4	MLP	0.0015	4.35e-04	0.9999	1.0000	0.0804	0.0341
	DBN	0.0011	4.21e-04	0.9999	1.0000	0.0405	0.0392
5	MLP	0.0023	4.95e-04	0.9998	0.9999	0.0748	0.0586
	DBN	0.0022	3.51e-04	0.9998	1.0000	0.0497	0.0189
6	MLP	0.0056	0.0020	0.9989	0.9994	0.2262	0.0721
	DBN	0.0050	0.0014	0.9991	0.9997	0.0361	0.0284

sensors and the lack of any specific information in the datasheets, a uniform noise distribution has been selected. Then, different sequences of white noise, with the amplitude defined as a percentage of the maximum output level, have

been generated. The levels of 1%, 3%, and 5%, respectively, have been considered and added to the input signals of the networks.

In Fig. 15, the RMSE obtained by simulating the models in the presence of noise added to all the inputs is reported. As it can be observed from Fig. 15, the SS implemented by the DBN is more robust to noise with respect to the PCA + MLP model. The output 6 is more sensitive to noise for both the models.

To perform a deeper analysis, the noise has been added to each input at once. The results obtained with a 5% white noise, for both the PCA + MLP and the DBN, are shown in Figs. 16 and 17, respectively.

Furthermore, in Figs. 16 and 17, it can be observed that the SS implemented with the DBN is more robust to noisy measurements. Moreover, the sensors of the input array, which more affect the SS behavior, are those denoted as inputs 1 and 2, i.e., TGS6810 and TGS825 in the case of the DBN SS, respectively.

V. CONCLUSION

In this article, a gas monitoring system for industrial environments is proposed. It is based on a gas sensor array, whose outputs are further elaborated by a deep neural network, obtaining an SS with a good accuracy. Two approaches are compared in this article. The first one was based on the PCA and a classical MLP. It was introduced to exploit the high linear correlation among the inputs. This is due to the low selectivity of solid-state gas sensors toward gas targets. In this way, we can deembend the common mode of the sensor array, reaching the right balance between a number of inputs and hardware specifications. This solution showed a good performance and allowed a sensible reduction of the parameters in the model. However, this configuration not revealed an adequate robustness to noisy measurements. A second approach is based on DBN, due to the reliability to better approximate complex functions. This second design leads to a sensible reduction of the number of model parameters and revealed a better performance toward noisy measurements. The parameter reduction can bring to a simpler implementation on a microcontroller-based embedded board. It could be equipped with a specific shield designed for the chosen sensors. This system could conjugate the microcontroller features with personal computer ones in the same environment, and it could be tuned for the specific application. A similar hardware configuration is reported in [33].

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